ICE-4107: Microwave Engineering

Lecture 2

Department of Information & Communication Technology

Bangladesh University of Professionals

Introduction

- **Electron Beam** is a stream of electrons**.** Free electrons in a vacuum can be manipulated by electric and magnetic fields to form a fine beam. Where the beam collides with solid-state matter, electrons are converted into heat or kinetic energy. This concentration of energy in a small volume of matter can be precisely controlled electronically, which brings many advantages.
- The electron beam's velocity is believed to be in a uniform electric field, uniform magnetic field, or uniform electromagnetic field.
- All current microwave devices use a uniform field to interact with electrons.
- Our primary purpose here is to provide with a background for understanding the **electron-field interactions** in microwave devices.

Electron Motion in an Electric Field

- To understand fields and electron-field interactions, certain experimental laws of electricity and magnetism are important.

- The fundamental force law of charges is Coulomb's law states that:

"Between two charges there exists either an attractive or a repulsive force, depending on whether the charges are of opposite or like sign."

If we express in terms of equations,

$$\mathbf{F} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2}\mathbf{u}_{R_{12}} \qquad \text{newtons} \qquad (1\text{-}1\text{-}1)$$

Electron Motion in an Electric Field

F/m = Farad/meter

where Q = charge in coulombs

$\epsilon_0 = 8.854 \times 10^{-12} \approx \frac{1}{36\pi} \times 10^{-9}$ F/m is the permittivity of free space

R = range between the charges in meters

$\mathbf{u}$ = unit vector

The electric field intensity produced by the charges is defined as the force per unit charge—that is,

$$\mathbf{E} \equiv \frac{\mathbf{F}}{Q} = \frac{Q}{4\pi\epsilon_0 R^2}\mathbf{u}_R \qquad \text{V/m} \qquad (1\text{-}1\text{-}2)$$

Electron Motion in an Electric Field

If there are n charges, the electric field becomes

$$\mathbf{E} = \sum_{m=1}^{n} \frac{Q_m}{4\pi\epsilon_0 R_m^2} \mathbf{u}_{R_m} \qquad (1\text{-}1\text{-}3)$$

To determine the path of an electron in an electric field, the force must be related to the mass and acceleration of the electron by Newton's second law of motion. So

$$\mathbf{F} = -e\mathbf{E} = m\boldsymbol{a} = m\frac{dv}{dt} \qquad (1\text{-}1\text{-}4)$$

where $m = 9.109 \times 10^{-31}$ kg, mass of electron
$\boldsymbol{a}$ = acceleration in meters per second squared
v = velocity of electron in meters per second
$e = 1.602 \times 10^{-19}$ C, charge of electron that is negative

Electron Motion in an Electric Field

- The force is in the opposite direction of the field because the electron has a negative charge.
- When an electron moves in an electric field **E**, it experiences a force -e**E** newtons. The differential equations of motion for an electron in an electric field in rectangular coordinates are given by

$$\frac{d^2x}{dt^2} = -\frac{e}{m}E_x \qquad (1\text{-}1\text{-}5a)$$

$$\frac{d^2y}{dt^2} = -\frac{e}{m}E_y \qquad (1\text{-}1\text{-}5b)$$

$$\frac{d^2z}{dt^2} = -\frac{e}{m}E_z \qquad (1\text{-}1\text{-}5c)$$

where $e/m = 1.759 \times 10^{11}$ C/kg is the ratio of charge to mass of electron and E_x, E_y, E_z are the components of **E** in rectangular coordinates.

Electron Motion in an Electric Field

- The cylindrical coordinates (r, ϕ, z) are defined as in Fig. 1-1-1. It can be seen that

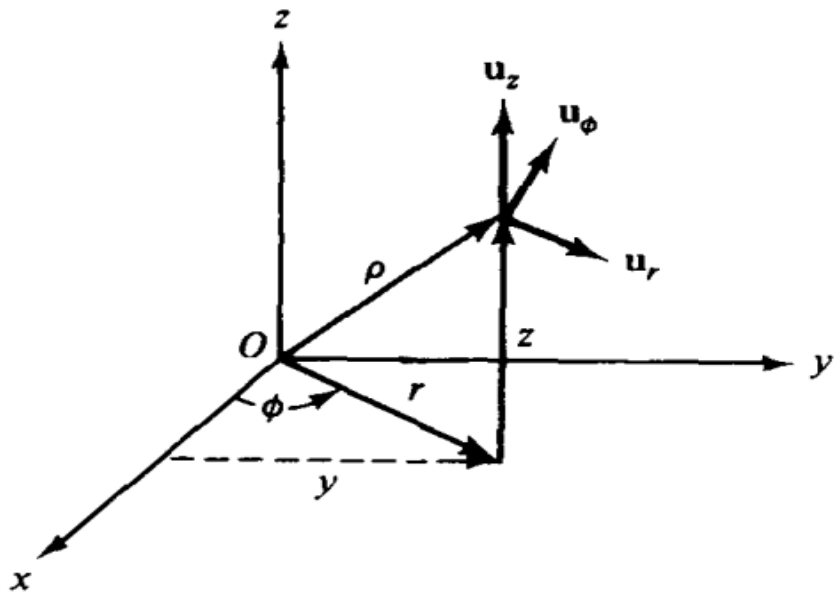


Figure 1-1-1 Cylindrical coordinates.

$$x = r\cos\phi \qquad (1\text{-}1\text{-}6a)$$

$$y = r\sin\phi \qquad (1\text{-}1\text{-}6b)$$

$$z = z \qquad (1\text{-}1\text{-}6c)$$

Electron Motion in an Electric Field

$\varphi = Phi$

and, conversely,

$$r = (x^2 + y^2)^{1/2} \tag{1-1-7a}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) = \sin^{-1}\frac{y}{(x^2 + y^2)^{1/2}} = \cos^{-1}\frac{x}{(x^2 + y^2)^{1/2}} \tag{1-1-7b}$$

$$z = z \tag{1-1-7c}$$

A system of unit vectors, $\mathbf{u}_r$, $\mathbf{u}_\varphi$, $\mathbf{u}_z$, in the directions of increasing r, φ, z, respectively, is also shown in the same diagram. While $\mathbf{u}_z$ is constant, $\mathbf{u}_r$ and $\mathbf{u}_\varphi$ are functions of ϕ ; that

$$\mathbf{u}_r = \cos\phi\,\mathbf{u}_x + \sin\phi\,\mathbf{u}_y \tag{1-1-8a}$$

$$\mathbf{u}_\phi = -\sin\phi\,\mathbf{u}_x + \cos\phi\,\mathbf{u}_y \tag{1-1-8b}$$

Electron Motion in an Electric Field

Differentiation of Eqs. (1-1-8) with respect to ϕ yields

$$\frac{d\mathbf{u}_r}{d\phi} = \mathbf{u}_\phi \tag{1-1-9a}$$

$$\frac{d\mathbf{u}_\phi}{d\phi} = -\mathbf{u}_r \tag{1-1-9b}$$

The position vector $\boldsymbol{\rho}$ can be expressed in cylindrical coordinates in the form

$$\boldsymbol{\rho} = r\mathbf{u}_r + z\mathbf{u}_z \tag{1-1-9c}$$

Differentiation of Eq. (1-1-9c) with respect to t once for velocity and twice for acceleration yields

$$\mathbf{v} = \frac{d\boldsymbol{\rho}}{dt} = \frac{dr}{dt}\mathbf{u}_r + r\frac{d\mathbf{u}_r}{dt} + \frac{dz}{dt}\mathbf{u}_z = \frac{dr}{dt}\mathbf{u}_r + r\frac{d\phi}{dt}\cdot\frac{d\mathbf{u}_r}{d\phi} + \frac{dz}{dt}\mathbf{u}_z$$

$$= \frac{dr}{dt}\mathbf{u}_r + r\frac{d\phi}{dt}\mathbf{u}_\phi + \frac{dz}{dt}\mathbf{u}_z \tag{1-1-10}$$

Electron Motion in an Electric Field

$$\boldsymbol{a} = \frac{d\mathbf{v}}{dt} = \left[\frac{d^2r}{dt^2} - r\left(\frac{d\phi}{dt}\right)^2\right]\mathbf{u}_r + \left(r\frac{d^2\phi}{dt^2} + 2\frac{dr}{dt}\frac{d\phi}{dt}\right)\mathbf{u}_\phi + \frac{d^2z}{dt^2}\mathbf{u}_z$$

$$= \left[\frac{d^2r}{dt^2} - r\left(\frac{d\phi}{dt}\right)^2\right]\mathbf{u}_r + \frac{1}{r}\frac{d}{dt}\left(r^2\frac{d\phi}{dt}\right)\mathbf{u}_\phi + \frac{d^2z}{dt^2}\mathbf{u}_z \qquad (1\text{-}1\text{-}11)$$

Therefore the equations of motion for electrons in an electric field in cylindrical coordinates are given by

$$\frac{d^2r}{dt^2} - r\left(\frac{d\phi}{dt}\right)^2 = -\frac{e}{m}E_r \qquad (1\text{-}1\text{-}12a)$$

$$\frac{1}{r}\frac{d}{dt}\left(r^2\frac{d\phi}{dt}\right) = -\frac{e}{m}E_\phi \qquad (1\text{-}1\text{-}12b)$$

$$\frac{d^2z}{dt^2} = -\frac{e}{m}E_z \qquad (1\text{-}1\text{-}12c)$$

where E_r, E_ϕ, and E_z are the components of $\mathbf{E}$ in cylindrical coordinates.

Electron Motion in a Magnetic Field

A charged particle in motion in a magnetic field of flux density **B** is experimentally found to experience a force that is directly proportional to charge *Q*, its velocity **v** and flux density **B** and the sine of the angle between the vectors v and **B** ,

We can write,

$$F = Q\mathbf{v} \times \mathbf{B}$$

Since the electron has negative charge

$$F = -e\mathbf{v} \times \mathbf{B}$$

Electron Motion in a Magnetic Field

The motion equations of an electron in a magnetic field in rectangular coordinates can be written as

$$\frac{d^2x}{dt^2} = -\frac{e}{m}\left(B_z\frac{dy}{dt} - B_y\frac{dz}{dt}\right) \qquad (1\text{-}2\text{-}3a)$$

$$\frac{d^2y}{dt^2} = -\frac{e}{m}\left(B_x\frac{dz}{dt} - B_z\frac{dx}{dt}\right) \qquad (1\text{-}2\text{-}3b)$$

$$\frac{d^2z}{dt^2} = -\frac{e}{m}\left(B_y\frac{dx}{dt} - B_x\frac{dy}{dt}\right) \qquad (1\text{-}2\text{-}3c)$$

Thank You